

Binomial expansion - Set 1

TMUA practice questions by topic. Specification MM2.4. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 Find the coefficient of x^3 in the expansion of $(2 + 3x)^5$.
- A 270
B 1080
C 2160
D 720
E 540
F 360
- 2 The coefficient of x^2 in the expansion of $(1 + kx)^6$ is 60. Given that $k > 0$, find k .
- A 2
B 4
C 3
D $\sqrt{10}$
E $\frac{1}{2}$
F 10
- 3 What is the sum of the coefficients in the expansion of $(3x - 2)^7$?
- A 2187
B -128
C 0
D 1
E -1
F 128

- 4 Evaluate $\binom{10}{3} - \binom{10}{7} + \binom{9}{2}$.
- A 36
B 240
C 0
D 84
E 120
F 156
- 5 In the expansion of $(1+x)^n$, the coefficients of x^4 and x^5 are equal. Find n .
- A 20
B 5
C 11
D 9
E 10
F 8
- 6 Expand $(1-2x)^4$ and hence find the coefficient of x^2 in $(1+3x)(1-2x)^4$.
- A 24
B -24
C 48
D 0
E 12
F -48
- 7 Find the coefficient of x^5 in the expansion of $(1+x)^3(1+x)^4$.
- A 35
B 21
C 12
D 1
E 28
F 7

- 8** Use the binomial expansion to find the value of 1.01^5 correct to 4 decimal places.
- A** 1.0515
B 1.0510
C 1.0505
D 1.0500
E 1.0410
F 1.0501
- 9** Find the term independent of x in the expansion of $\left(x^2 + \frac{2}{x}\right)^6$.
- A** 60
B 15
C 160
D 480
E 240
F 120
- 10** The first three terms in the expansion of $(1 + ax)^n$ in ascending powers of x are $1 + 12x + 54x^2$. Find n .
- A** 9
B 6
C 10
D 4
E 12
F 8

11 Which of the following statements is/are true?

I $\binom{n}{r} = \binom{n}{n-r}$ for all integers $0 \leq r \leq n$

II $\binom{n}{r} + \binom{n}{r+1} = \binom{n+1}{r+1}$ for all integers $0 \leq r < n$

III $\sum_{r=0}^n \binom{n}{r} = 2^n$ for all positive integers n

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

12 Find the value of $(\sqrt{2} + 1)^4 + (\sqrt{2} - 1)^4$.

A 34

B 36

C 17

D $34 + 24\sqrt{2}$

E $24\sqrt{2}$

F 16